

LINKED LISTS & TREES

COMPUTER SCIENCE MENTORS 61A

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1 Linked Lists

For the following questions, use this dataclass-based linked list. A `LinkedList[T]` is either a `Link[T]` or the empty tuple `()`. The type variable `T` describes the element type.

```
from __future__ import annotations
from dataclasses import dataclass

type LinkedList[T] = Link[T] | tuple[()]

@dataclass
class Link[T]:
    first: T
    rest: LinkedList[T] = ()

    def __str__(self) -> str:
        return format_link(self)

def format_link[T](s: Link[T]) -> str:
    string = '(' + str(s.first)
    remaining = s.rest
    while isinstance(remaining, Link):
        string += ' ' + str(remaining.first)
        remaining = remaining.rest
    assert remaining == (), 'Not a valid linked list'
    return string + ')'
```

Construct a list with `Link(1, Link(2))`. Read its elements with `first` and `rest`. At the end, `rest` is `()`, which has no `first` or `rest` attributes. `print` uses parentheses around the elements; entering a `Link` at the prompt shows the dataclass representation with field names.

1. Write `append`, which takes two linked lists and returns a new linked list with all of the elements of the first followed by all of the elements of the second. Then use `append` to write `reverse`, which returns a new linked list with the elements in reverse order.

```
def append[T](a: LinkedList[T], b: LinkedList[T]) -> LinkedList[T]:  
    """
```

```
    >>> print(append(Link(1, Link(2)), Link(7, Link(8))))
```

```
    (1 2 7 8)
```

```
    >>> print(append((), Link(5)))
```

```
    (5)
```

```
    >>> print(append(Link(1), ()))
```

```
    (1)
```

```
    >>> append((), ())
```

```
    ()
```

```
    """
```

```
    if _____:
```

```
        return _____
```

```
    return _____
```

```
def reverse[T](lnk: LinkedList[T]) -> LinkedList[T]:
```

```
    """
```

```
    >>> print(reverse(Link(1, Link(2, Link(3)))))
```

```
    (3 2 1)
```

```
    >>> reverse(())
```

```
    ()
```

```
    """
```

```
    if _____:
```

```
        return _____
```

```
    return _____
```

2. Write `keep_if`, which takes a linked list of integers `lnk` and a predicate `pred`, and returns a new linked list of only the elements for which `pred` returns true, in their original order.

```
from collections.abc import Callable
def keep_if(
    lnk: LinkedList[int], pred: Callable[[int], bool]
) -> LinkedList[int]:
    """
    >>> lnk = Link(1, Link(2, Link(3, Link(4))))
    >>> print(keep_if(lnk, lambda x: x % 2 == 0))
    (2 4)
    >>> print(keep_if(lnk, lambda x: x > 0))
    (1 2 3 4)
    >>> keep_if(lnk, lambda x: x > 10)
    ()
    """
    if _____:
        return _____

    rest = _____

    if _____:
        return _____

    return _____
```

3. Write `merge`, which takes two linked lists of integers, each sorted in increasing order, and returns a new sorted linked list containing all of the elements of both.

```
def merge(a: LinkedList[int], b: LinkedList[int]) -> LinkedList[int]:
    """
    >>> a = Link(1, Link(5))
    >>> b = Link(2, Link(3, Link(8)))
    >>> print(merge(a, b))
    (1 2 3 5 8)
    >>> print(merge(a, ()))
    (1 5)
    >>> print(merge(Link(1, Link(3)), Link(1)))
    (1 1 3)
    >>> merge((), ())
    ()
    """
    if _____:

        return _____

    elif _____:

        return _____

    elif _____:

        return _____

    else:

        return _____
```

2 Trees

For the following questions, use this dataclass-based tree. A `Tree[T]` has a label of type `T` and a list of branches, each of which is also a `Tree[T]`. A tree is never empty.

```
from __future__ import annotations
from dataclasses import dataclass, field

@dataclass
class Tree[T]:
    label: T
    branches: list[Tree[T]] = field(default_factory=list) # defaults to []

    def __str__(self) -> str:
        return format_tree(self)

def is_leaf(t: Tree) -> bool:
    return not t.branches

def format_tree(t: Tree, indent: int = 0) -> str:
    s = '  ' * indent + str(t.label)
    for b in t.branches:
        s += '\n' + format_tree(b, indent + 1)
    return s
```

4. What would Python display? Run these statements in order. If a statement causes an error, identify the error and continue.

```
>>> t = Tree(5, [Tree(3, [Tree(8)]), Tree(1), Tree(7)])
>>> t.label
```

```
>>> t.branches[0].label
```

```
>>> is_leaf(t.branches[1])
```

```
>>> len(t.branches)
```

```
>>> t.branches[1].branches[0]
```

```
>>> t.branches[0].branches
```

```
>>> u = Tree(2, [t.branches[0]])  
>>> u.branches[0].branches[0].label
```

```
>>> print(u)
```

5. Implement `tree_height`, which takes in a tree and returns its height: the depth of the lowest leaf, where the root has depth 0.

```
def tree_height(t: Tree[int]) -> int:  
    """  
    >>> tree_height(Tree(1))  
    0  
    >>> t = Tree(1, [Tree(4), Tree(2, [Tree(3)])])  
    >>> tree_height(t)  
    2  
    """  
    if _____:  
        return _____  
  
    return 1 + _____
```

6. Write `all_paths`, which takes a tree and returns a list of all root-to-leaf paths. Each path is a linked list of the labels from the root down to a leaf.

```
def all_paths[T](t: Tree[T]) -> list[LinkedList[T]]:
    """
    >>> t = Tree(1, [Tree(2), Tree(3, [Tree(4), Tree(5)])])
    >>> for p in all_paths(t):
    ...     print(p)
    (1 2)
    (1 3 4)
    (1 3 5)
    >>> all_paths(Tree(7))
    [Link(first=7, rest=())]
    """
    if _____:

        return _____

    return _____
```

7. Write `count_paths`, which takes a tree `t` of integers and an integer `total`, and returns the number of root-to-leaf paths whose labels sum to `total`.

```
def count_paths(t: Tree[int], total: int) -> int:
```

```
    """
```

```
    >>> t = Tree(1, [Tree(3),
...                 Tree(3, [Tree(4),
...                         Tree(0)]),
...                 Tree(4, [Tree(1)])
...               ])
    ...    )
```

```
    >>> count_paths(t, 4)
```

```
    2
```

```
    >>> count_paths(t, 8)
```

```
    1
```

```
    >>> count_paths(t, 6)
```

```
    1
```

```
    >>> count_paths(t, 5)
```

```
    0
```

```
    """
```

```
    if _____:
```

```
        return _____
```

```
    return _____
```